

Recognise that the real number system includes the rational numbers and the irrational numbers, and solve problems involving real numbers using digital tools

Learning intention

- recognise that the real number system includes the rational numbers and the irrational numbers, and solve problems involving real numbers using digital tools.

Teach and model

- using positive and negative rational numbers to solve problems.
- investigating the position of rational and irrational numbers on the real number line.
- using a real number line to indicate the solution interval for inequalities of the form $ax + b < c$.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.